Identification of an Unknown – “A”

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Introduction

The complete chemical structure of an unknown compound can be determined by evaluating various spectroscopies: Mass Spectroscopy (GCMS), Nuclear Magnetic Resonance (NMR), and the Fourier Transform Infrared Spectroscopy (FTIR). By analyzing GCMS, the molecular structure of the compound can be determined. The complexity of the compound can be further analyzed by consulting the NMR, specifically by obtaining the structural formula. Lastly, FTIR can be analyzed to understand the types of chemical bonds (sigma, pi) and the functional groups found in the compound. This is a necessary skill to learn, as it is important to understand that an identity of a substance can be determined purely through empirical data.

Theory/Body:

Mass Spectroscopy (GCMS):

In any mass spectrometer, an unknown compound enters the ion source where it changes to a gas phase. By sorting the gas ions by their mass-to-charge (m/z) ratios, the corresponding electric current reflects the number of ions in the compound. This ratio is particularly important because it typically signifies the molecular weight of the unknown compound. While there are numerous types of mass spectrometer, a gas chromatograph-mass spectrometer (GCMS) will be analyzed for this lab report. The sample from the gas chromatograph directly enters the mass spectrometer, so that each compound/atom in the mixture is analyzed. Therefore, the number of each atom of the compound and its corresponding identity can be determined through this instrument. The result of GC-MS is a mass spectrum data in a histogram graph, where the horizontal x-axis represents the mass-to-charge (m/z) ratio and the vertical y-axis represents the ion intensity of the compound.

In order to determine the basic molecular formula of the unknown compound, the Rule of Thirteen is first evaluated. Therefore, the starting hydrocarbon formula is obtained. By analyzing the GC-MS graph and identifying the molecular ion peak and the surrounding peaks (M+1 and M+2, respectively), one can identify whether there are additional atoms in the compound, such as oxygen, nitrogen, chlorine, and bromine. For example, by recognizing that the height of M+1 and M+2 peaks are nearly identical, one can observe that there is a bromine atom in the unknown compound. The corresponding hydrocarbon, molecular weight equivalent of bromine can then be subtracted from the starting hydrocarbon formula. This way, the molecular structure of the compound can be determined.

Nuclear Magnetic Resonance (NMR):

The nuclear magnetic resonance (NMR) instrumentation is useful in further understanding the complexity of the compound. More specifically, the structural formula of the unknown compound can be obtained through graphical analysis. The unknown sample is first irradiated with radio frequency energy to excite the hydrogen or carbon nuclei. This excitation of the nucleus produces an oscillating curve that represents the structure of the compound.

There are four important types of information that can be obtained when reading the NMR graph. First, one can determine the different types of protons: primary, secondary, and tertiary. Primary hydrogens represent a carbon with three hydrogens, secondary hydrogens represent a carbon with two hydrogens, and tertiary hydrogens represent a carbon with one hydrogen. Secondly, one can determine the abundance of these different types of protons and how frequently they occur in the compound. Additionally, one can understand the connectivity of the different atoms in the compound through a concept called ‘chemical shift.’ Chemical shift explains that the NMR graph explains the more complex parts of the unknown compound towards the left-hand side of the graph. Lastly, one can identify connectivity and locations of the atoms by understanding the multiplicity of lines, which is known as ‘spin-spin coupling’.

Fourier-Transform Infrared Spectroscopy (FTIR):

Lastly, one can further understand the complexity of the unknown compound by analyzing the Fourier transform infrared spectroscopy (FTIR). More specifically, one can identify the type of vibration, such as C-H stretching or -CH_2 bending, that occurs in the compound. Additionally, one can understand the different bond orders, or the frequency of the Bonds between two atoms (e.g., C-C, O-H, N-H). Therefore, this useful instrumental device can allow one to fully grasp where different functional groups are located as well as where aromatic parts are located in the compound.

FTIR are analyzed by identifying the characteristic absorption peaks of the FTIR graph. By determining its intensity, such strong, moderate, or weak, one can refer to a table (Table III-1) that shows the types of stretching/bending and bond orders at specific positions of the graph. For example, if there are strong peaks located from 1300 cm^{-1} to 1000 cm^{-1}, this signifies that there are C-O stretches in esters. This way, the complete analysis and identification of the compound can be determined, and the IUPAC name can be consequently labeled.

Data

See Attachment 1.

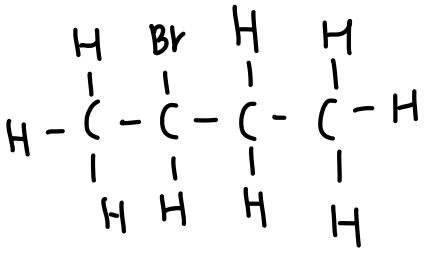
Analysis:

Gas Chromatography Mass Spectroscopy (GCMS):

For the unknown sample "A", the molecular ion peak is observed to be at m/z 136. Because this is an even value, it signifies that the unknown compound contains no or an even number of nitrogen atoms. Located on the right of the M peak, there also appears to be a M+2 peak of similar abundance. Because the ratio of M/(M+2) is close to 1:1, this indicates that the compound also contains a bromine atom. The Rule of Thirteen can be applied to evaluate the molecular formula of the unknown sample. First, the M peak value can be divided by 13, such that 136/13=10 with a remainder of 6. If the compound were a hydrocarbon, the molecular formula would be: $C_{10}H_{16}$. However, a bromine atom must be included in this molecular formula. The carbon equivalent of Br is C_6H_7. Therefore, the possible molecular formula may be $C_{10}H_{16}$ - C_6H_7 + Br = C_4H_9Br.

Nuclear Magnetic Resonance (NMR):

Through GCMS, the molecular formula was determined to be C_4H_9Br. Through Nuclear Magnetic Resonance (NMR), the structural formula can be determined such that the locations and connectivity of the atoms can be evaluated. More specifically, the multiplicity of protons can be determined: 3H, 3H, 2H, and 1H (evaluated from the right-hand-side of the graph to the left-hand-side of the graph). Primary hydrogens (1H) are represented by -CH. Secondary hydrogens (2H) are represented by -CH_2. Lastly, tertiary hydrogens (3H) are represented by -CH_3. By fitting these molecular parts together and adding a bromine to a primary hydrogen, the structural formula can be obtained:



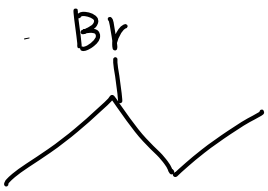
Fourier-Transform Infrared Spectroscopy (FTIR):

Lastly, Fourier-Transform Infrared Spectroscopy (FTIR) can be analyzed to further understand the chemical structure of the compound by determining the chemical bonds and/or functional groups of the compound. For this unknown sample, FTIR was evaluated to confirm the chemical structure that was obtained through GCMS and NMR. From the FTIR graph, medium to weak infrared (IR) peaks are observed at 2971.47 cm^{-1} and settle down at ~2750 cm^{-1}. Consulting Table III-1. In Appendix III, this vibration range of 3100-3000 cm^{-1} represents a C-H stretch. Furthermore, IR peaks are observed from 1456.32 cm^{-1} and 1380.13 cm^{-1}, which represent a C-C stretch. IR peaks from 785 cm^{-1} to 540 cm^{-1} represent an alkyl halide, or C-Br. Clearly, FTIR can be used to confirm the molecular structure of compound "A", such as its functional groups (Br) and chemical bonds (single sigma bonds).

Conclusion

Based on the analysis of the three spectroscopies (GCMS, NMR, and FTIR), the unknown compound "A" was determined to be 2-Bromobutane. First, GCMS was consulted to obtain the simple, molecular structure of the compound. Thereafter, NMR was consulted to obtain the structural structure of the compound, thereby determining the connectivity of the atoms. Lastly, FTIR was consulted to confirm the chemical bonds and functional groups of the compound.

Chemical Structure:



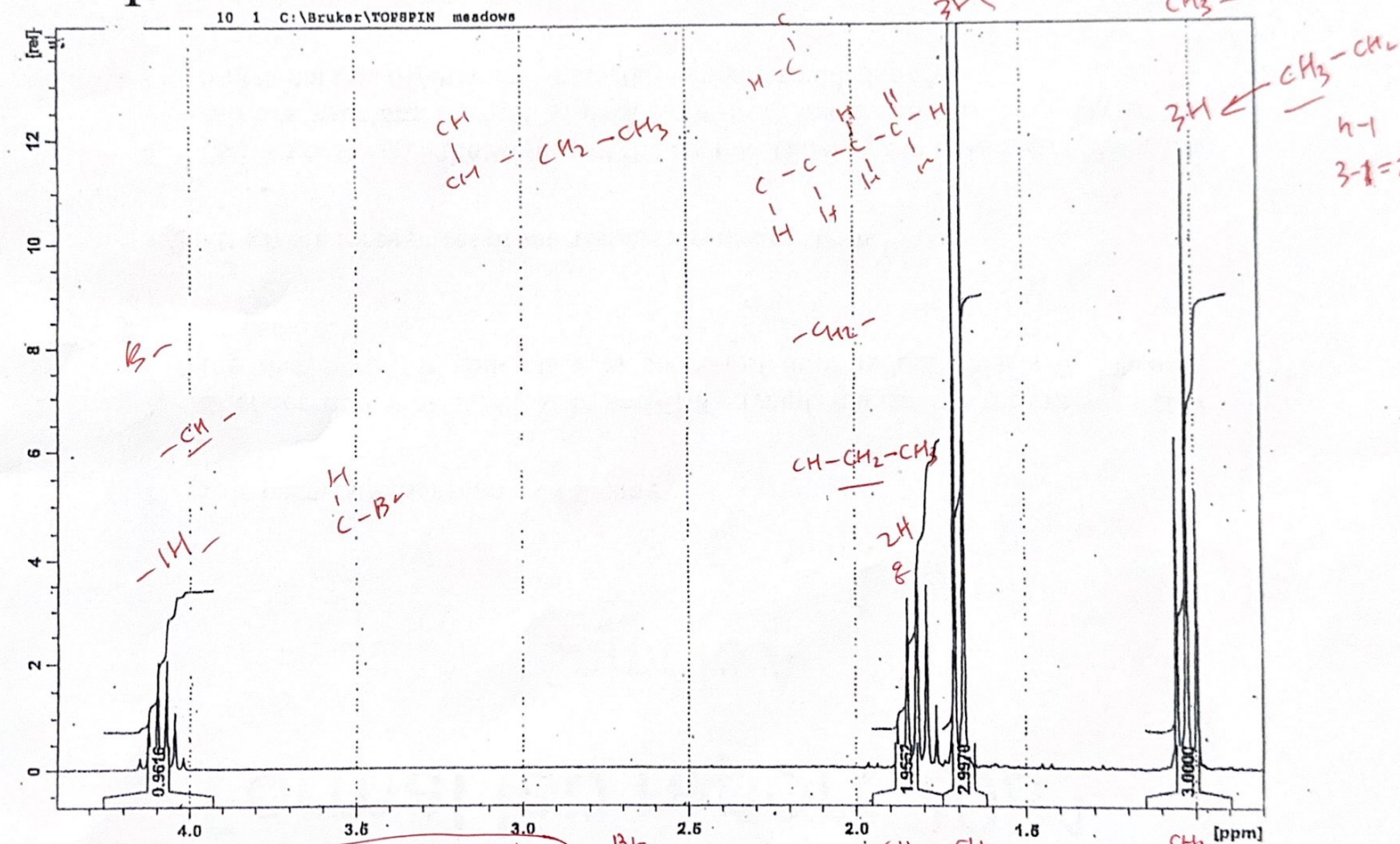
2-Bromobutane

References

Keller L, Joseph K, and Miguel S. 2023 Fall. CHEM 2081 Organic Chemistry II Lab Manual [lab manual]. Auburn (AL): Auburn University.

NMR Sample A

Sample A



9+1=10

C_4H_9Br

C_nH_{2n+2}

n=4

2n+2

2(4)+2=10

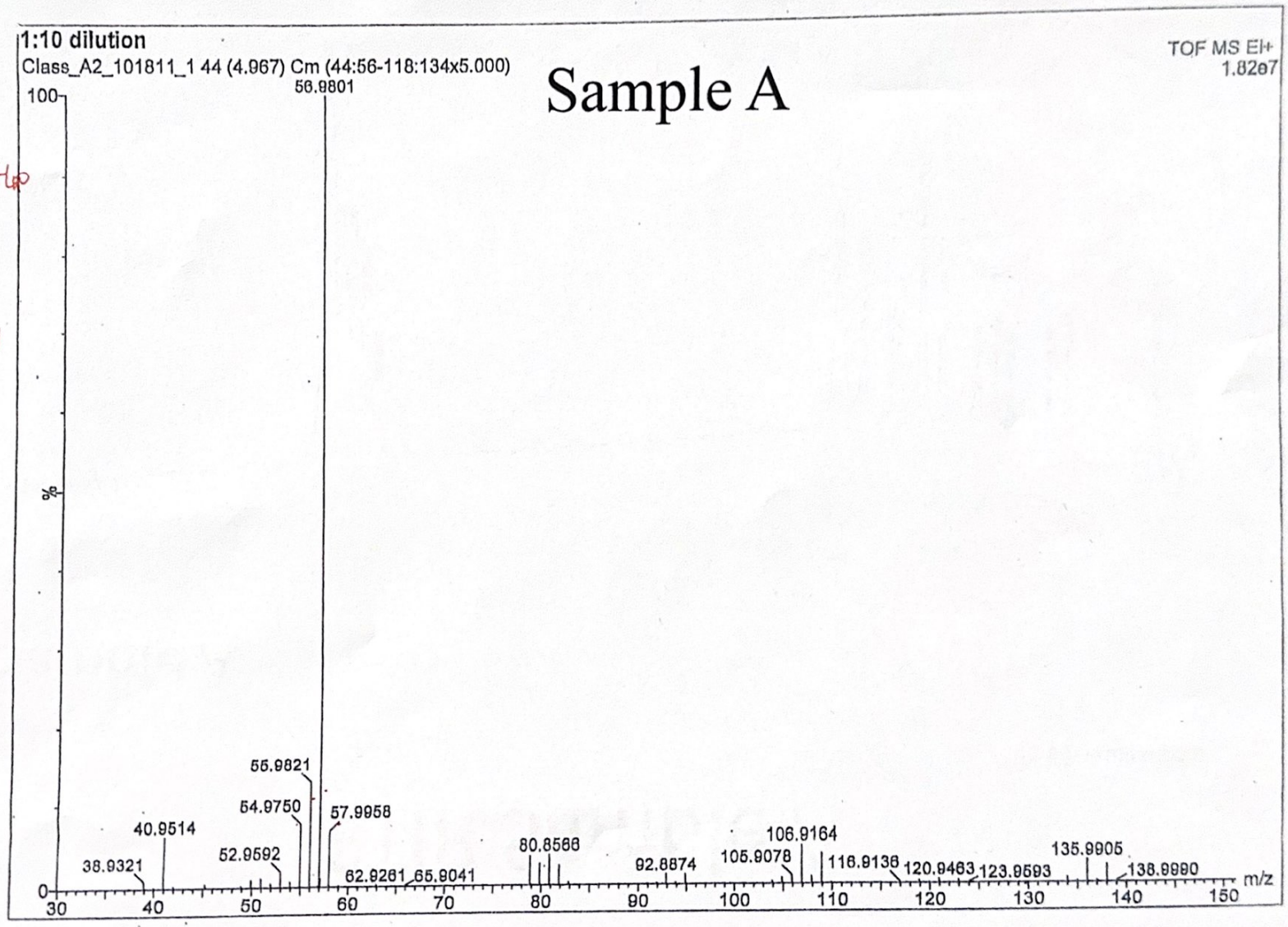
$C_4H_{10} - C_4H_{10}$

=0

no pi,
no rings

no carbonyl
=O

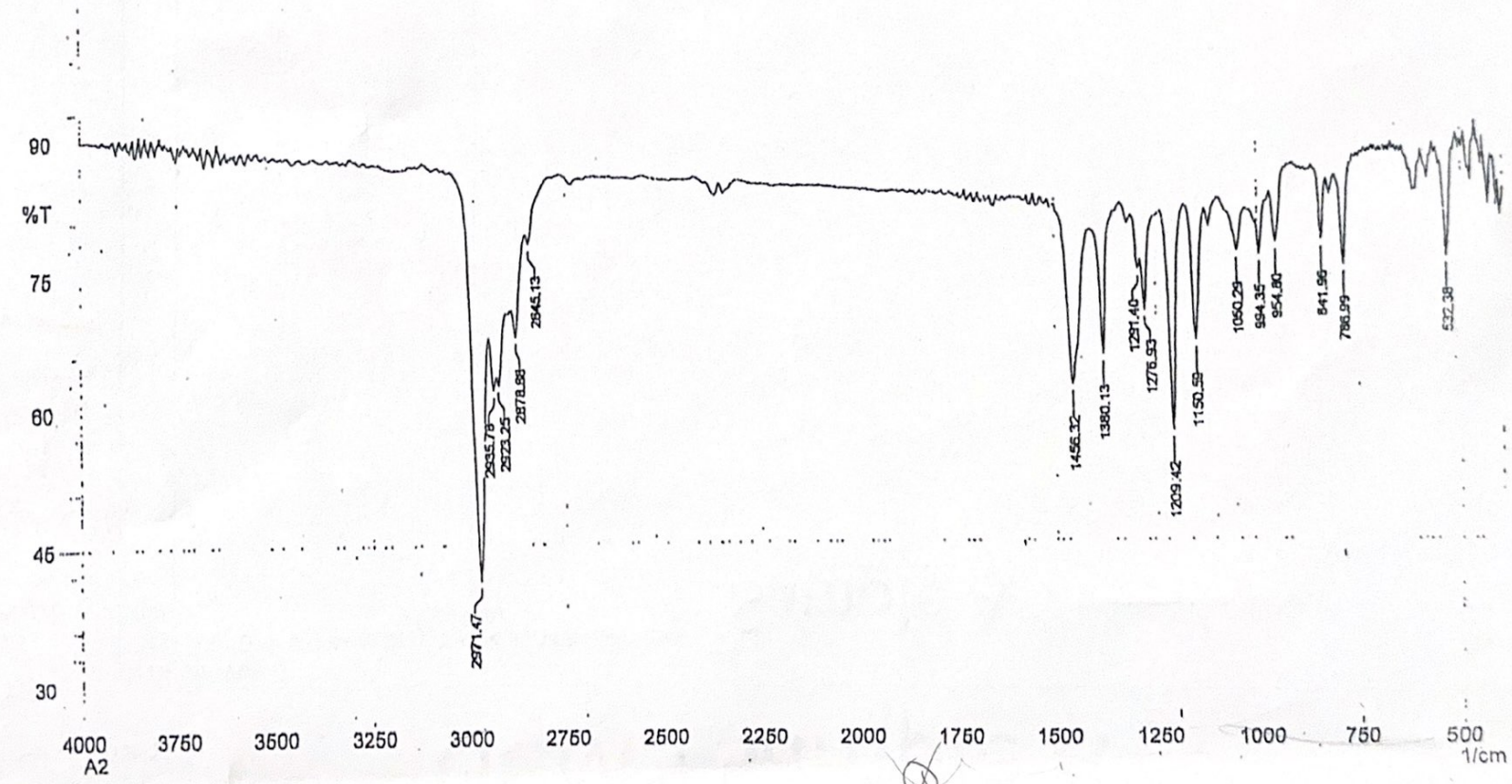
GCMS Sample A



FTIR Sample A

SHIMADZU

Sample A



Comment;
A2

No. of Scans;
Resolution;
Apodization;

Date/Time; 3/2/2011 2:24:12 PM
User; User